Auditing Social Bias in Language Models: Metrics & a Category-Aware Framework

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1 Research Question and Contributions

How do fine-tuning and prompt engineering, alone and combined, change bias in BERT-family content moderation classifiers, measured as counterfactual sensitivity beyond paraphrase-induced variation?

We make three contributions. (1) A controlled comparison of encoder-only fine-tuning and prompt engineering, alone and combined, on the same base model and data, isolating each intervention's effect on moderation bias. (2) Applying the baseline-controlled counterfactual metric of Yang et al. (2026) to content moderation, using multiple paraphrased alternates per post. (3) A reusable auditing framework that accepts both encoder classifiers and LLMs, with a working prototype. We scope the study to one BERT-family classifier, base and fine-tuned, with gender and disability as the bias axes; LLM fine-tuning and dialect are out of scope.

2 Literature Review

Content moderation classifiers often confuse toxic intent with the mere mention of a demographic group: Dixon et al. (2018) found that they flag benign comments simply for containing identity terms. Because Blodgett et al. (2020) urge bias research to state precisely which harm it studies, we target an allocational harm: comments with the same message receiving different toxicity scores solely because of demographic words.

Counterfactual evaluation. Garg et al. (2019) used demographic substitutions to measure and reduce classifier bias, and Czarnowska et al. (2021) formalized such tests as score changes under demographic perturbation. A confound remains: every counterfactual edit also changes surface wording, so a raw score shift conflates targeted bias with general prompt sensitivity. Yang et al. (2026) show that gender swaps flip predictions about as often as simple paraphrases, and propose testing targeted effects against a paraphrase baseline matched to the edit length, using per-sample distributional metrics (JSD) and significance tests. We adopt this metric for content moderation.

Limits of prior audits. Gardner et al. (2021) argue that balancing data for isolated words does not rule out reliance on biased feature combinations. CAFFE (Parziale et al., 2025) proposes structured, intent-aware test cases with conversational context and reports broader bias coverage than metamorphic testing. Pseudo-log-likelihood audits on CrowS-Pairs (Nangia et al., 2020) are also fragile: Liu (2024) shows that counting which sentence scores higher ignores the score distribution and lacks robustness on limited data. Nor do pre-training likelihoods reliably predict fine-tuned classifier behavior.

Datasets. We use resources by role. Civil Comments (Borkan et al., 2019) provides identity-annotated toxicity labels for training, and Toxic-Commons (Arnett et al., 2024) adds texts classified across five dimensions, including gender/sex-based and ability-based discrimination. ETHOS (Mollas et al., 2020) contributes hate speech annotations on YouTube and Reddit comments.

2.1 Research Gap

Most counterfactual evaluations swap a single word and focus on gender or race. We address two gaps.

Multi-word swaps. Because existing metrics typically swap one word at a time, they can miss identity markers that depend on surrounding context, qualifiers, and grammatical dependencies, and may therefore underestimate bias on real-world text. We instead swap all co-referring words together, including descriptive modifiers (e.g., *she/woman/her* to *he/man/his*), and compare against controls that change the same number of words.

Disability. Most toxicity studies focus on gender or race, leaving disability largely overlooked. This is an important blind spot. Hutchinson et al. (2020) and Venkit et al. (2023) showed that classifiers regularly inflate toxicity scores on harmless comments simply because they mention disability terms (such as *blind/wheelchair*). Benchmarks like BBQ (Parrish et al., 2022) similarly confirm that models default to negative stereotypes in ambiguous contexts, frequently penalizing respectful discussions of disability while failing to catch actual ableist harm (Phutane et al., 2025).

3 Data and Approach

Data. We use cleaned subsets of the ETHOS Hate Speech (Mollas et al., 2020) and ToxicCommons (Arnett et al., 2024). All testing and counterfactual perturbations are conducted on strictly held-out splits from these datasets.

Data tagging. An 8B LLM tags each post with the words to swap along each bias axis and the synonyms to swap for controls. To validate it, we manually verify 200 tagged posts and have a frontier model (Gemini 3.8 Flash) verify another 200 with an overlap of 100, yielding 300 verified posts and a measure of human-model agreement.

Training. We fully fine-tune a pretrained modern BERT-family classifier (Model 0 - M0) on cleaned social media posts to produce M2, then add prompt engineering to each, yielding M1 and M3 (Figure 1). Each variant's instruction prompt specifies output format and asks if the post is toxic; prompt engineering adds labeled few-shot examples.

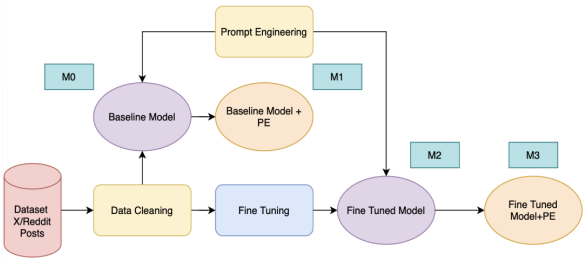


Figure 1: Training pipeline: one dataset, two interventions, four variants (M0 to M3).

Testing. The framework takes a set of models and a dataset and returns a comparative bias analysis (Figure 2). For each post, it generates a swap along a bias axis (e.g., he/she for gender) and several controls that change the same number of words through neutral synonym swaps. Since

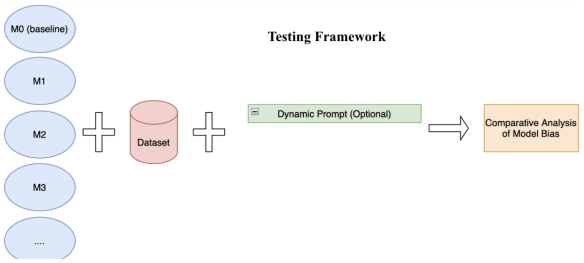


Figure 2: Testing framework: model variants are audited on the same dataset, with an optional prompt.

models react to any wording change (Yang et al. (2026)), we define bias as $\Delta = \mathrm{JSD}(P_x \,\|\, P_{x^t}) - \frac{1}{K}\sum_{k=1}^{K} \mathrm{JSD}(P_x \,\|\, P_{x^{(k)}})$, where P_x, P_{x^t}, and $P_{x^{(k)}}$ are the model's output distributions for the original post, the targeted swap, and the k-th of K controls. A model counts as biased only if Δ is consistently positive across posts (one-sided paired t-test). We also report a signed direction statistic (whether the version referencing the targeted group, e.g., women or overweight people, scores as more toxic) and pairwise model comparisons.

Proof of concept. Our prototype, a notebook backed by source modules, ran end-to-end on two public toxicity classifiers over 10 posts (Figure 3). Both moved more under neutral rewording than under the gender swap (mean Δ of -0.0008 and -0.0032 bits), and no direction or between-model effect was significant. At $n = 10$ this is expected: the POC shows the comparison works, not a finding about either model.

Contribution Statement

For the research question and scope: Shreyas Natarajan
For system design: Shreyas Natarajan, Gayathri Ethiraj, Dharini Venkatesan
For the literature review: Gayathri Ethiraj, Dharini Venkatesan
For the dataset selection and curation: Gayathri Ethiraj, Dharini Venkatesan
For the approach and data tagging: Shreyas Natarajan
For the proof of concept code: Shreyas Natarajan

AI Use Disclosure

Shreyas Natarajan: Claude code help for the proof of concept, Claude for formatting for overleaf
Gayathri Ethiraj: ChatGPT and Claude for factual checks, paraphrasing and formatting for overleaf
Dharini Venkatesan: Gemini for factual checks,

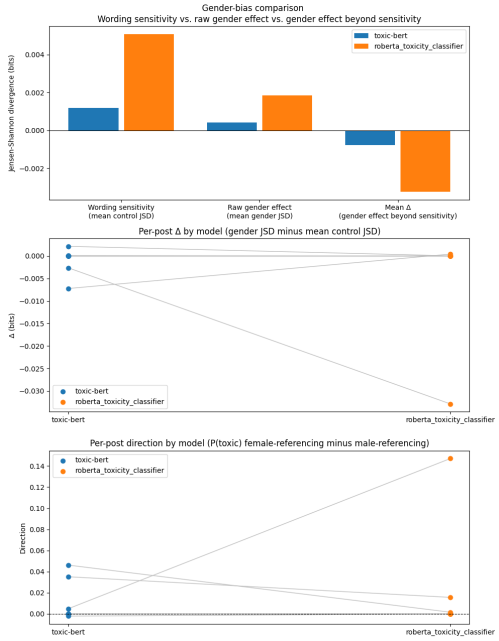


Figure 3: POC results. Top: wording sensitivity, raw gender effect, and mean Δ. Middle: per-post Δ. Bottom: per-post direction.

paraphrasing and formatting for overleaf

References

Catherine Arnett, Eliot Jones, Ivan P. Yamshchikov, and Pierre-Carl Langlais. 2024. Toxicity of the Commons: Curating open-source pre-training data. arXiv preprint arXiv:2410.22587. https://arxiv.org/abs/2410.22587

Soumya Barikeri, Anne Lauscher, Ivan Vulić, and Goran Glavaš. 2021. RedditBias: A real-world resource for bias evaluation and debiasing of conversational language models. In *Proceedings of ACL*. https://aclanthology.org/2021.acl-long.151/

Su Lin Blodgett, Solon Barocas, Hal Daumé III, and Hanna Wallach. 2020. Language (technology) is power: A critical survey of "bias" in NLP. In *Proceedings of ACL*. https://arxiv.org/abs/2005.14050

Daniel Borkan, Lucas Dixon, Jeffrey Sorensen, Nithum Thain, and Lucy Vasserman. 2019. Nuanced metrics for measuring unintended bias with real data for text classification. In *Companion Proceedings of The Web Conference*. https://arxiv.org/abs/1903.04561

Paula Czarnowska, Yogarshi Vyas, and Kashif Shah. 2021. Quantifying social biases in NLP: A generalization and empirical comparison of extrinsic fairness metrics. *Transactions of the ACL*. https://arxiv.org/abs/2106.14574

Lucas Dixon, John Li, Jeffrey Sorensen, Nithum Thain, and Lucy Vasserman. 2018. Measuring and mitigating unintended bias in text classification. In *Proceedings of AIES*. https://doi.org/10.1145/3278721.3278729

Matt Gardner, William Merrill, Jesse Dodge, Matthew E. Peters, Alexis Ross, Sameer Singh, and Noah A. Smith. 2021. Competency problems: On finding and removing artifacts in language data. In *Proceedings of EMNLP*, pages 1801–1813. https://aclanthology.org/2021.emnlp-main.135/

Sahaj Garg, Vincent Perot, Nicole Limtiaco, Ankur Taly, Ed H. Chi, and Alex Beutel. 2019. Counterfactual fairness in text classification through robustness. In *Proceedings of AIES*. https://arxiv.org/abs/1809.10610

Yang Liu. 2024. Robust evaluation measures for evaluating social biases in masked language models. In *Proceedings of AAAI*, 38(17):18707–18715. https://doi.org/10.1609/aaai.v38i17.29834

Ioannis Mollas, Zoe Chrysopoulou, Stamatis Karlos, and Grigorios Tsoumakas. 2020. ETHOS: An online hate speech detection dataset. arXiv preprint arXiv:2006.08328. https://arxiv.org/abs/2006.08328

Nikita Nangia, Clara Vania, Rasika Bhalerao, and Samuel R. Bowman. 2020. CrowS-Pairs: A challenge dataset for measuring social biases in masked language models. In *Proceedings of EMNLP*. https://arxiv.org/abs/2010.00133

Alessandra Parziale, Gianmario Voria, Valeria Pontillo, Gemma Catolino, Andrea De Lucia, and Fabio Palomba. 2025. Toward systematic counterfactual fairness evaluation of large language models: The CAFFE framework. arXiv preprint arXiv:2512.16816. https://arxiv.org/abs/2512.16816

Eric Michael Smith, Melissa Hall, Melanie Kambadur, Eleonora Presani, and Adina Williams. 2022. "I'm sorry to hear that": Finding new biases in language models with a holistic descriptor dataset. In *Proceedings of EMNLP*, pages 9180–9211. https://arxiv.org/abs/2205.09209

Zihao Yang, Mosh Levy, Yoav Goldberg, and Byron C. Wallace. 2026. Compared to what? Baselines and metrics for counterfactual prompting. arXiv preprint arXiv:2605.01048. https://arxiv.org/abs/2605.01048

Ben Hutchinson, Vinodkumar Prabhakaran, Emily Denton, Kellie Webster, Yu Zhong, and Stephen Denuyl. 2020. Social biases in NLP models as barriers for persons with disabilities. In *Proceedings of ACL*. https://aclanthology.org/2020.acl-main.487/

Pranav Narayanan Venkit, Mukund Srinath, and Shomir Wilson. 2023. Automated ableism: An exploration

of explicit disability biases in sentiment and toxicity analysis models. In *Proceedings of the Workshop on Trustworthy NLP (TrustNLP)*. https://aclanthology.org/2023.trustnlp-1.16/

Mahika Phutane, Ananya Seelam, and Aditya Vashistha. 2025. "Cold, calculated, and condescending": How AI identifies and explains ableism compared to disabled people. In *Proceedings of ACM FAccT*. https://arxiv.org/abs/2410.03448

Alicia Parrish, Angelica Chen, Nikita Nangia, Vishakh Padmakumar, Jason Phang, Jana Thompson, Phu Mon Htut, and Samuel R. Bowman. 2022. BBQ: A hand-built bias benchmark for question answering. In *Findings of ACL*. https://aclanthology.org/2022.findings-acl.165/

This proposal is available on Overleaf.[1]

[1] https://www.overleaf.com/5121283976stdnzsxhgtbw#d7345f